This assignment will not be graded and is only for practice.

Level 1:

1) Consider a function $f : \mathrm{R}^2 \to \mathrm{R}$ defined as:

$$f(x,y) = \begin{cases} x^2 \sin\frac{1}{x} + y & \text{if } x = 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Statement 1: f_y is continuous at the origin.
Statement 2: f_x is continuous at the origin.

Enter the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

2) A function $f : \mathrm{R}^2 \to \mathrm{R}$ satisfies that $f_x(x,y) = c$ and $f_y(x,y) = k$ of all points of R^2, where c and k are some constants, then which of the following option(s) is (are) true? ***1 point***

- ☐ The tangent line exists at any point of R^2 in any direction.
- ☐ The tangent lines at the origin do not lie on a plane.
- ☐ There exist points in R^2 for which tangent points in all directions do not exist.
- ☐ $f(x,y)$ can be $f(x,y) = cx + ky + \alpha$, where α is any real number.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The tangent line exists at any point of R^2 in any direction.
$f(x,y)$ can be $f(x,y) = cx + ky + \alpha$, where α is any real number.

3) Consider $f : \mathrm{R}^2 \to \mathrm{R}$ and suppose partial derivatives of $f(x,y)$ with respect to x and y are continuous at a point (a,b). Then which of the following option(s) is (are) true? ***1 point***

- [ ] $f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y)$
- [ ] $f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$
- [ ] $f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y).$
- [ ] $f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y)$
$f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$

4) Consider a function $f(x,y) = x^2 + y^2$. Which of following is the tangent at the point (1,2) in the direction of (1, 1)? ***1 point***

- ○ $x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$
- ○ $x = 1, y(t) = 2 + t, z(t) = 5 + \frac{6t}{\sqrt{2}}$
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$

Level 2:

5) Consider a function **1 point**

$$f(x,y) = \begin{cases} \frac{x}{y} & \text{if } y = 0 \\ 0 & \text{if } y = 0. \end{cases}.$$

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line $x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}}$?

- $x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t$
- $x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$
- $x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$
- $x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$

6) Consider a function $f(x,y,z) = 2x^2 + xy + z^2$. Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0) **1 point**

- $(1, 1, 1, 4 + 5t)$
- $(1 + t, 1, 1, 4 + t)$
- $(1, 1 + t, 1 + t, 4 + 5t)$
- $(1 + t, 1, 1, 4 + 5t)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1 + t, 1, 1, 4 + 5t)$

7) **1 point**

Consider the following functions f defined as:

$$f(x, y, z) = xyz.$$

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$
- $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$
- None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$

Your score is: 0/7